Problem 2

The length of the minute hand of a certain clock is $R = 1.2$ m, and the length of the hour hand is $r = 1$ m. What are the angular velocities of the hands, and what are the speeds of their tips?

Solution

Angular Velocities

The angular velocity of a rotating object is given by:

$$\omega = \frac{2\pi}{T},$$

where T is the period of rotation.

For the minute hand, $T = 60$ minutes $= 3600$ s:

$$\omega_{\mathrm{m}} = \frac{2\pi}{3600} \approx 1.75 \times 10^{-3} \mathrm{rad/s}.$$

For the hour hand, $T = 12$ hours $= 43200$ s:

$$\omega_{\mathrm{h}} = \frac{2\pi}{43200} \approx 1.45 \times 10^{-4} \mathrm{rad/s}.$$

Linear Speeds

The linear speed of the tip of a hand is given by:

$$v = \omega R.$$

For the minute hand:

$$v_{\mathrm{m}} = \omega_{\mathrm{m}} R = (1.75 \times 10^{-3})(1.2) \approx 2.1 \times 10^{-3} \mathrm{m/s}.$$

For the hour hand:

$$v_{\mathrm{h}} = \omega_{\mathrm{h}} r = (1.45 \times 10^{-4})(1) \approx 1.45 \times 10^{-4} \mathrm{m/s}.$$

Final Answer

- Angular velocity of the minute hand: $\omega_{\mathrm{m}} \approx 1.75 \times 10^{-3} \mathrm{rad/s}$.
- Angular velocity of the hour hand: $\omega_{\mathrm{h}} \approx 1.45 \times 10^{-4} \mathrm{rad/s}$.
- Speed of the tip of the minute hand: $v_{\mathrm{m}} \approx 2.1 \times 10^{-3} \mathrm{m/s}$.
- Speed of the tip of the hour hand: $v_{\mathrm{h}} \approx 1.45 \times 10^{-4} \mathrm{m/s}$.